Sentiment Classification on GoEmotions: A Comparative Study of Text Classification Methods

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***Abstract*—In this work, we study sentiment classification on the GoEmotions dataset by mapping fine-grained emotions to four coarse sentiment categories: positive, negative, neutral, and ambiguous. To formulate the problem as a single-label classification task, we retain only samples with exactly one emotion label. We train and compare two classification models: a Term Frequency-Inverse Document Frequency (TF-IDF) feature pipeline with a Logistic Regression classifier, and a fine-tuned DistilBERT transformer. Our experimental results show that the TF-IDF + Logistic Regression baseline achieves a classification accuracy of 0.636 (95% Confidence Interval (CI) [0.622, 0.650]) and macro-F1 of 0.601 (95% CI [0.585, 0.616]). Fine-tuning DistilBERT on the full training dataset yields a test accuracy of 0.715 (95% CI [0.703, 0.728]) and macro-F1 of 0.684 (95% CI [0.669, 0.699]). We report 95% bootstrap confidence intervals for both models, demonstrating the range of performance variability on the test set.**

Introduction

Emotion and sentiment analysis are fundamental tasks in Natural Language Processing (NLP). These tasks enable the conversion of raw, human language into structured data denoting emotional intent, making them useful techniques for a wide range of applications, including customer service, social media analysis, and public opinion monitoring. The widespread adoption of social media has especially led to a massive influx of informal text, which is often characterized by noisy writing, slang, and complex emotional expressions. Standard sentiment analysis (classifying text into positive, negative, or neutral) is often insufficient for capturing the nuances of these diverse expressions.

The GoEmotions dataset (Demszky et al. 2020) is a large, human-annotated corpus with approximately 58,000 Reddit comments labeled with 28 fine-grained emotion categories. In this project, we address a simplified formulation of the task: 4-way single-label sentiment classification. We filter the comments to keep only those with exactly one emotion label, and map that label to one of four sentiment classes: positive, negative, neutral, or ambiguous.

To address this task, we compare two modeling approaches. The first is a baseline model that combines TF-IDF features derived from word-level n-grams with a Logistic Regression classifier. The second is a transformer-based approach that fine-tunes a pre-trained DistilBERT model on the training data.

We evaluate both models on the test split, report their classification accuracy and macro-F1 scores along with 95% bootstrap confidence intervals. We also analyze their per-class performance to determine the best-performing model for the task.

Problem Description

The sentiment classification task is framed as a supervised single-label classification problem. Given an input text string $x \in \mathcal{X}$ representing a single Reddit comment, the objective is to predict a label $y \in \mathcal{Y}$. Prior to feature extraction, the text is preprocessed by collapsing consecutive whitespace, removing newline characters, and stripping leading and trailing whitespace. The label space is defined as

$$\mathcal{Y} = \{\text{negative}, \text{neutral}, \text{positive}, \text{ambiguous}\}$$

Sentiment Mapping

We map the original 28 GoEmotions categories to our target sentiment classes as follows:

- **Positive:** admiration, amusement, approval, caring, desire, excitement, gratitude, joy, love, optimism, pride, relief.
- **Negative:** anger, annoyance, disappointment, disapproval, disgust, embarrassment, fear, grief, nervousness, remorse, sadness.
- **Ambiguous:** confusion, curiosity, realization, surprise.
- **Neutral:** neutral.

To formulate the problem as a single-label classification task, we retain only comments with exactly one emotion label.

Methodology

We compare a linear baseline with a transformer-based model for sentiment classification.

TF-IDF + Logistic Regression (Baseline)

The baseline model maps raw comments into a sparse vector space using term frequency-inverse document frequency (TF-IDF) feature weighting. Word-level n-grams (unigrams and bigrams) are extracted to capture both explicit emotional descriptors and local word combinations (such as negation phrases) (Pedregosa et al. 2011). The resulting vectors are passed to a multi-class Logistic Regression classifier, which models the probability of each sentiment class as a linear combination of the lexical features. To prevent overfitting to the large feature space, the model is trained with L2 regularization, and class weights are adjusted to compensate for training set imbalance.

Fine-Tuned DistilBERT (Transformer)

DistilBERT (Sanh et al. 2019) is a compact transformer model derived from BERT (Devlin et al. 2019) via knowledge distillation to learn deep contextualized text representations. Unlike TF-IDF, which treats words independently of their position, DistilBERT utilizes self-attention mechanisms to dynamically compute representations where a word's vector changes depending on its surrounding tokens. Input text is segmented using WordPiece subword tokenization to mitigate out-of-vocabulary terms. The token sequence is then processed through transformer layers to capture long-range semantic and syntactic dependencies. For classification, the final hidden state corresponding to the special `[CLS]` token is used as a sequence-level representation. A linear classification layer maps this contextual vector to logits corresponding to the target sentiment classes. During training, the pre-trained encoder weights and the classification head are jointly fine-tuned on our target data.

Experimental Setup

This section details the dataset partitions, model hyperparameters, and metrics used in our evaluation.

Dataset Splits

After filtering multi-label comments and mapping the original 28 emotions into four sentiment categories, the resulting dataset is split into training, validation, and test sets. The class counts for each split are summarized in Table I.

Table I: Dataset splits and class counts after single-label filtering.

Sentiment Class	Train	Val	Test	Total
Positive	12,920	1,668	1,603	16,191
Neutral	12,823	1,592	1,606	16,021
Negative	7,012	853	932	8,797
Ambiguous	3,553	435	449	4,437
Total	36,308	4,548	4,590	45,446

Hyperparameters & Training Settings

To ensure reproducibility, all dataset shuffling, model initializations, and evaluations are fixed with a random seed of 42.

For the TF-IDF + Logistic Regression baseline, features are computed using word-level n-grams with $n \in \{1, 2\}$ (Pedregosa et al. 2011). We restrict the vocabulary to the top 50,000 features, filter out features appearing in only one document (min_df = 2), and apply sublinear term-frequency scaling ($\text{tf} \leftarrow 1+\log(\text{tf})$). The Logistic Regression classifier is regularized with an inverse strength parameter of $C = 1.0$ and optimized using the L-BFGS solver with a maximum of 1,000 iterations. The model uses balanced class weights to compensate for class imbalance.

For DistilBERT, input texts are padded or truncated to a maximum sequence length of 64 tokens. The model is fine-tuned for two epochs using the AdamW optimizer with a learning rate of 2×10^{-5} and a batch size of 16. Training was conducted using GPU acceleration. Unlike the baseline, DistilBERT is trained using standard cross-entropy loss without class weight adjustment.

Evaluation Metrics

We evaluate both models on the held-out test split using:

- **Classification Accuracy:** The percentage of comments predicted correctly.
- **Macro-F1 Score:** The unweighted average of F1-scores across all sentiment classes, ensuring that minority classes (such as "ambiguous") are weighted equally.
- **Per-Class F1-Score:** To evaluate performance variations across specific sentiments.

To quantify uncertainty and capture the range of performance variability on the test set, we estimate 95% confidence intervals (CI) for both accuracy and macro-F1 using bootstrapping with $N = 1,000$ resamples.

Results and Discussion

Overall Performance

Table II summarizes the overall performance of the two models on the test set.

Table II: Overall test performance of the classification models.

Model	Accuracy (95% CI)	Macro-F1 (95% CI)
TF-IDF + LogReg	0.636 [0.622, 0.650]	0.601 [0.585, 0.616]
DistilBERT	**0.715** [0.703, 0.728]	**0.684** [0.669, 0.699]

DistilBERT outperforms the TF-IDF baseline, improving accuracy from 0.636 to 0.715 (+7.9 percentage points) and macro-F1 from 0.601 to 0.684 (+8.3 percentage points). The 95% bootstrap confidence intervals represent the uncertainty under resampling, demonstrating that DistilBERT

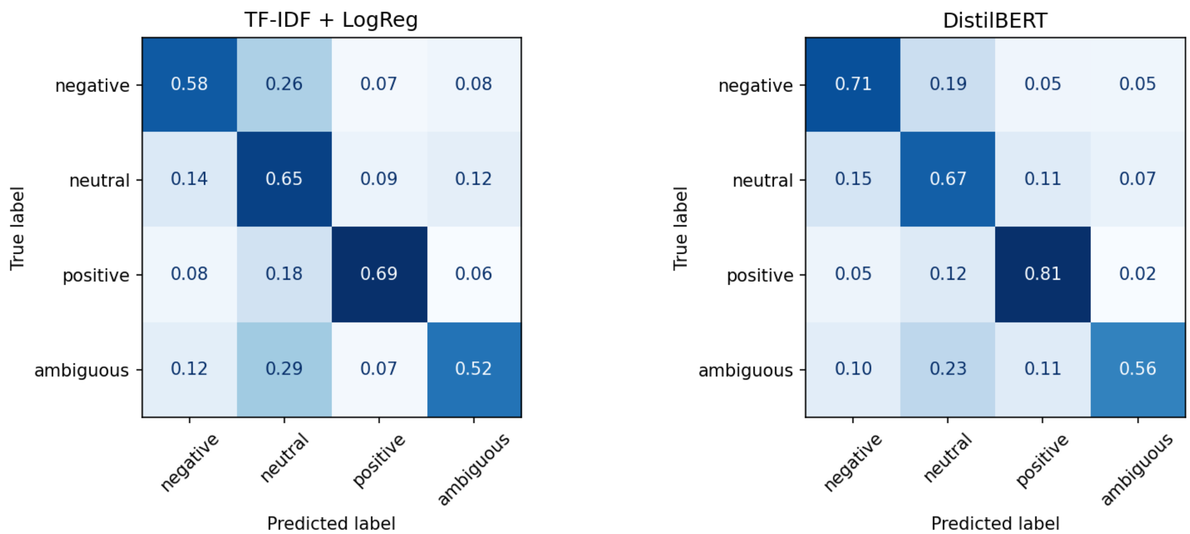

Figure 1: Normalized confusion matrices for TF-IDF + Logistic Regression (left) and fine-tuned DistilBERT (right).

consistently operates in a higher performance band on this test set.

Per-Class F1 Analysis

Table III reports the F1-score for each sentiment class.

Table III: Per-class F1-scores on the test set.

Class	TF-IDF + LogReg	DistilBERT	Diff
negative	0.576	**0.674**	+0.098
neutral	0.629	**0.679**	+0.050
positive	0.747	**0.818**	+0.071
ambiguous	0.452	**0.566**	+0.114

DistilBERT achieves higher F1-scores for every sentiment class. The largest improvements occur for the ambiguous (+0.114) and negative (+0.098) classes, suggesting that contextual transformer representations are particularly effective for capturing indirect, nuanced, and context-dependent emotional expressions. More modest gains are observed for the positive (+0.071) and neutral (+0.050) classes, indicating that these categories are comparatively easier for both models to identify.

Visualizations

Fig. 2 shows the overall test accuracy with 95% bootstrap confidence intervals, Fig. 3 compares per-class F1-scores, and Fig. 1 presents the normalized confusion matrices for both models.

Quantitative and Qualitative Error Analysis

To understand the limitations of both modeling approaches, we conduct an error analysis on their classification failures on the test split. First, we compute the quantitative distribution of errors across the true sentiment classes, reported in Table IV.

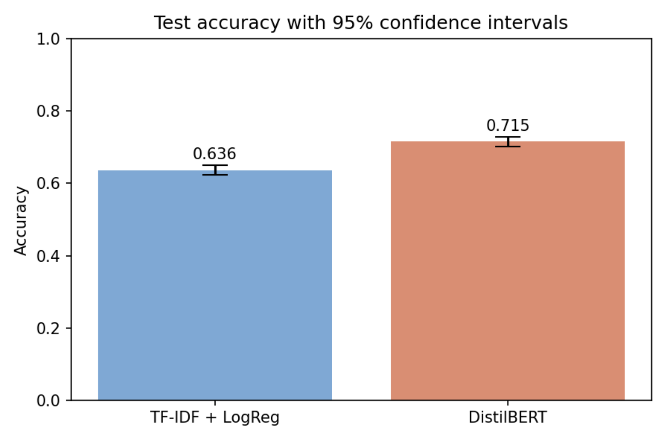

Figure 2: Classification accuracy with 95% bootstrap confidence intervals.

The quantitative breakdown in Table IV reveals that both models exhibit the highest error rates on the ambiguous class (47.7% for the baseline and 44.3% for DistilBERT), indicating that non-polar or curiosity-driven sentiments remain challenging. DistilBERT achieves substantial error reductions on the polar classes: it reduces the error rate on positive comments by 12.2 percentage points and on negative comments by 12.6 percentage points. However, the error rate reduction on the neutral class is minimal (reducing by only 2.1 percentage points). Consequently, neutral comments constitute the largest share of DistilBERT's errors (40.9%).

TF-IDF + Logistic Regression Errors:

- **Inability to Capture Valence Shifts and Negation:** Because the baseline relies on sparse n-gram occurrences, it fails to capture contextual shifts in

Table IV: Error rates and error distributions by true class on the test set.

	TF-IDF + Logistic Regression		DistilBERT	
True Class	Error Rate	% of Total Errors	Error Rate	% of Total Errors
positive	31.2%	29.9%	19.0%	23.3%
negative	41.5%	23.2%	28.9%	20.6%
neutral	35.4%	34.1%	33.3%	40.9%
ambiguous	47.7%	12.8%	44.3%	15.2%
Total	36.4%	100.0%	28.5%	100.0%

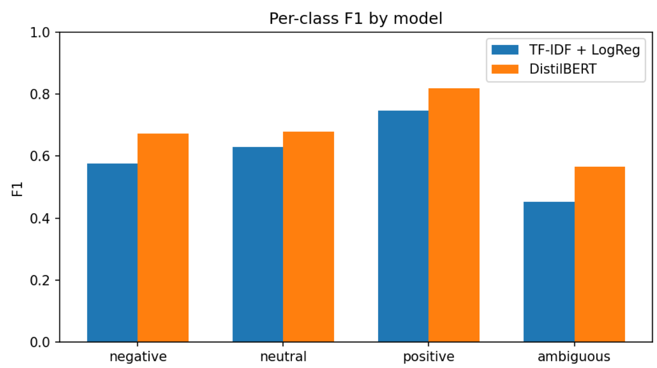

Figure 3: Per-class F1-scores on the test set.

sentiment. For example, it misclassified the negative comment *"I'm sorry to hear that friend :(. It's for the best. . . "* as *positive* due to the high coefficients of words like `"best"`, `"friend"`, and `"accept"`. Similarly, it misclassified *"It's wonderful because it's awful."* (which is *positive*) as *negative* due to the presence of the word `"awful"`.

- **Sensitivity to Superficial Lexical Cues:** The baseline is highly sensitive to explicit lexical indicators. Features with large positive-class coefficients, such as `"lol"` (10.28) and `"thanks"` (10.18), strongly bias the model toward *positive* predictions, even in sarcastic contexts. Conversely, since the *neutral* class lacks distinct emotional vocabulary, the model is forced to rely on noisy structural features like `"they"` (2.31) or bot commands like `"remindme"` (1.55), which generalize poorly.
- **Vulnerability to Out-of-Vocabulary (OOV) Terms:** The word-level representation cannot generalize to spelling variations, typos, or slang unseen during training.

DistilBERT Errors:

- **Sensitivity to Phrasing Style in Neutral Text:** DistilBERT tends to over-predict the *negative* class when comments contain strong punctuation or demanding phrasing, even if the content itself is neutral. For example, it misclassified *"You take that back right now! [NAME] does not deserve this."* (labeled *neutral* in GoEmotions) as *negative*.
- **Reduced Performance on Short, Elliptical Queries:** For short, elliptical queries, DistilBERT may fail to capture meaning when structural cues dominate. For instance, for the ambiguous question *"out of everyone why AC?"*, DistilBERT predicted *neutral*, whereas the baseline correctly identified it as *ambiguous* by matching the key feature `"why"`.
- **Lack of Model Interpretability:** Unlike the logistic regression baseline where the feature coefficients directly explain why a classification was made, DistilBERT's deep representations act as a black box, making individual failures difficult to diagnose or audit.

Computational Cost Comparison

The performance gains achieved by the transformer model come with substantial computational costs. The TF-IDF + Logistic Regression pipeline trains in under 1 second on a standard CPU and has negligible inference latency. In contrast, fine-tuning DistilBERT requires approximately 5 minutes on an NVIDIA GPU and exhibits significantly higher memory usage and inference latency. This represents a clear trade-off between classification performance and operational resource budgets.

Limitations

This study has several scoping limitations:

1. **Dataset Filtering:** By retaining only comments with exactly one emotion label, we discarded approximately half of the original GoEmotions dataset. This simplifies the multi-label nature of real-world emotion classification and biases the distribution toward unambiguous expressions.
2. **Short Sequence Length:** Truncating inputs to 64 tokens may discard context from longer comments.
3. **Domain Bias:** The models were trained and evaluated exclusively on Reddit comments, meaning they may not generalize well to other domains (e.g., formal news text or clinical notes) due to the prevalence of platform-specific slang and structures.

Ethical Considerations

The GoEmotions dataset is curated from public Reddit forums, which frequently contain offensive language, hate speech, or private user identifiers (although usernames are masked). Models trained on this data risk learning and propagating toxic biases or offensive speech patterns. In addition, automated sentiment classification of public text must be deployed with care to prevent misuse in invasive monitoring or mass surveillance.

Conclusion

These findings demonstrate that contextual transformer models substantially improve sentiment classification of noisy social media text. While TF-IDF remains valuable when efficiency and interpretability are priorities, DistilBERT provides a clear advantage for modeling context-dependent emotional expressions.

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